Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/09/2004
PI Name:	Albert, Mitchell S Ph.D.		
Project Title:	MRI Pulse Sequence Development		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mitchell.albert@umassmed.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	508-856-1096
Organization Name:	University of Massachusetts Medical School		
PI Address 1:	Department of Radiology		
PI Address 2:	55 Lake Avenue North		
PI Web Page:			
City:	Worcester	**State:**	MA
Zip Code:	01655	**Congressional District:**	3
Comments:	Note: PI moved to U Mass Medical School in 2007; previously at Brigham and Women's Hospital/Harvard University (9/2008)		
Project Type:	Ground	**Solicitation / Funding Source:**	PECASE AWARD
Start Date:	09/01/2003	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We are developing a new diagnostic technology: magnetic resonance imaging (MRI) and spectroscopy of hyperpolarized noble gases. Large non-equilibrium nuclear spin polarizations can be created in the spin-1/2 noble gases, 3He and 129Xe, using the technique of spin-exchange optical pumping. Such large polarizations greatly enhance the magnetic resonance detection of 3He and 129Xe, enabling high-resolution MRI studies of the lung and airways, as well as dissolved tissue-phase investigations of organs and tissues, such as the brain, using lipophilic 129Xe. An important feature of hyperpolarized noble gas MR is that it does not require a large magnetic field, unlike the conventional water proton MR method used in medical MRI. Low magnetic field MRI with hyperpolarized noble gases can allow us to use a small, light-weight, low-power devices. This capability provides potential for advances in both portable ground-based and practical space-based MR systems.		

	Our proposed ground-based investigations with hyperpolarized noble gas MR will affect biomedical research in two important ways: (1) by providing the potential to acquire novel information about physiological structure and function of the lung and the brain in space-flight and in terrestrial gravity, and (2) by developing a new diagnostic technology that has clinical applications for diagnosis of pathology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An important feature of hyperpolarized noble gas MR is that it does not require a large magnetic field, unlike the conventional thermally polarized water proton MR method used in medical MRI. As a result, at low magnetic fields (0.01 Tesla) the signal-to-noise ratio (SNR) of hyperpolarized noble gas MR is much larger than the SNR of conventional thermally polarized MR. Low field MR utilizing hyperpolarized noble gases should be practical in a small, light-weight, low-power device, providing the potential for both portable ground-based and practical space-based MR systems. Specific magnetic and biological properties of hyperpolarized 3He and 129Xe MRI include: (i) airway and lung gas space imaging (for diagnosis of chronic obstructive pulmonary disease (COPD), e.g. emphysema, asthma, and for studies of respiratory function); (ii) the imaging of lipid structures in the brain (for the diagnosis and staging of demyelinating diseases such as multiple sclerosis, and in research on brain function); and (iii) angiography and measurement of blood flow (for diagnosis of stroke and ischemia, and for research into the function of the brain and cardiopulmonary system).
Task Progress:	Adjustment of the TR (repeat time) parameter on the scans has been discovered to enable decreasing the static ventilation scan time from 10-13 seconds, down to 5-8 seconds. This gain cuts down on the time subjects need to hold their breath, which can be very helpful for patients with pulmonary diseases, vastly reducing their discomfort. A model-based approach has been developed to measure the diameters of airways captured with the dynamic imaging protocols. The rigor of this approach virtually eliminates the dependence on the unreliable human detection of airway edges, and the resultant randomness by which diameter endpoints are selected.
Bibliography Type:	Description: (Last Updated: 03/23/2012)